

Comparative anti-arthritic investigation of iridoid glycosides and crocetin derivatives from *Gardenia jasminoides* Ellis in Freund's complete adjuvant-induced arthritis in rats[☆]

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ABSTRACT

Background: Discovering novel compounds with higher activities is a key aim of natural products research. *Gardenia jasminoides* Ellis is a herb with anti-inflammatory properties. Iridoid glycosides (mainly geniposide) and crocetin derivatives (crocin) are the two major active constituents in this herb and are considered its active ingredients. However, which components are responsible for the anti-inflammatory properties of gardenia have remained to be investigated.

Purpose: Here, we prepared total iridoid glycosides (TIG) and total crocins (TC) from *G. jasminoides* Ellis, determined their main chemical constituents, and performed animal studies to evaluate their anti-adjuvant arthritis activities, thus, proposing a reasonable mechanism to explain the anti-inflammatory activities of the active components in this herbal remedy.

Study design: TIG and TC were prepared by using HPD-100 macroporous resin, and characterized by UHPLC-DAD-MS and UV-Vis spectrophotometer. Then, Freund's complete adjuvant-injected rats underwent drug treatments with TIG (160 mg/kg) and TC (160 mg/kg) for 14 days, and their ankle diameters were measured. Moreover, X-ray radiographs of the adjuvant injected hind paws were evaluated. Finally, histopathological examinations of the ankle joints, spleens and thymus were carried out to evaluate inflammatory reactions, and immunohistochemical measurements were conducted to evaluate TNF- α and TGF- β 1 expression in the ankle joint of the rats.

Results: The chemical composition determination of the current study showed that TIG was mainly composed of geniposide and TC was a fraction predominantly with crocin-1, crocin-2 and crocin-3. Calculation of results showed that TIG and TC contained 58.2% total iridoid glycosides and 54.7% total crocins, respectively. Our study suggested TIG and TC treatments markedly decreased paw swelling and ankle diameters of AA rats (both $p < 0.05$). The radiological analysis showed that administration of TIG and TC ameliorated bone destruction, and reduced the radiological bone destruction scores (TIG $p < 0.05$, TC $p > 0.05$). Moreover, data from histological assessment demonstrated considerable mitigation of inflammation in the joints (both $p < 0.01$), spleen and thymus of AA rats treated with TIG and TC. TNF- α and TGF- β 1 protein expression according to immunohistochemistry staining also supported the anti-arthritis activities of TIG and TC (TNF- α : TIG $p < 0.01$ and TC $p < 0.05$, TGF- β 1: TIG $p < 0.01$ and TC $p > 0.05$).

Conclusion: In the current study, fractionation of gardenia prior to further in vivo investigation has for the first time provided reasonable explanation for the anti-inflammatory activity of this herbal remedy. Our study

Abbreviations: AA, Adjuvant arthritis; FCA, Freund's complete adjuvant; H&E, hematoxylin-eosin; HSCCC, High-speed countercurrent chromatography; IHC, Immunohistochemistry; IOD, Integral optical density; RA, Rheumatoid arthritis; TC, total crocins; TCM, Traditional Chinese Medicine; TIG, total iridoid glycosides; TGF- β 1, Transforming growth factor beta1; TNF- α , tumor necrosis factor- α ; UHPLC-DAD-MS, Ultra high performance liquid chromatography-Diode array detector-Mass spectrometry; UV-Vis, Ultraviolet-visible

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showed that both TIG and TC from gardenia have anti-inflammatory properties. Overall, these experimental findings suggest that gardenia could be regarded as a potential therapeutic target for arthritis. However, as geniposide has a higher content than crocins in this herbal drug, TIG (mainly geniposide) seems to be primarily responsible for the anti-inflammatory properties of gardenia. Taken together, this maiden attempt demonstrated that TIG (mainly geniposide) is more important in evaluating the anti-inflammatory activity of *G. jasminoides* Ellis.

Introduction

Rheumatoid arthritis (RA) is a major chronic destructive disease worldwide and is characterized by joint swelling, synovial membrane inflammation, and cartilage and bone destruction. Although the pathogenesis of RA is still unknown, many researchers have confirmed that activated inflammatory T cells, B cells and macrophages invading the joint synovium are responsible for destruction of joint and cartilage (Al-Zifzaf et al., 2015; Hu et al., 2013). Moreover, various reports have demonstrated that macrophage-derived cytokines, predominantly interleukin (IL)-1 β , tumor necrosis factor (TNF- α) and transforming growth factor (TGF) β 1, are directly implicated in every phase of the inflammatory response associated with the pathogenesis of RA (McInnes and Schett, 2007; Pope, 2002; Xu et al., 2015). RA can rapidly progress to multi-system inflammation with irreversible joint destruction, thus increasing the risk of mortality (Jr, 1990). It has been reported that RA is one of the most common human autoimmune diseases with a prevalence of 0.5–1% in the adult population worldwide (Suzuki and Yamamoto, 2015). The life span of patients with RA is shortened due to the associated increase in cardiovascular and cerebrovascular diseases (Wolfe et al., 1994). Traditional non-steroidal antiinflammatory drugs (NSAIDs), disease-modifying antirheumatic drugs (DMARDs), glucocorticoids and novel biologics have been investigated and prescribed for the treatment of RA (Rathore et al., 2007; Smolen et al., 2007). However, these treatments, especially prolonged usage at high doses, are associated with some adverse effects such as hepatic and gastrointestinal disorders, immunodeficiency and humoral disturbance, and are costly and inconvenient (Roth, 2012; Scheinecker et al., 2008). In this regard, it is necessary to investigate the mechanism of RA and new therapeutic approaches. In particular, due to their efficiency, minimum adverse effects and low cost, increasing attention has been focused on ethnological medicinal plants for the development of new antirheumatic drugs for long-term administration (Liu et al., 2017).

G. jasminoides Ellis (Rubiaceae) is an evergreen shrub widely

distributed in tropical and subtropical regions, often found growing on mountain slopes or road sides as an ornamental plant. The dried ripe fruits of this plant have been recorded as Fructus Gardeniae (Chinese herbal name is “zhizi”) in the Chinese Pharmacopoeia and included in Traditional Chinese Medicine (TCM) formulations for diuretic, cholagogue, anti-inflammatory, and antipyretic effects (National Commission of Chinese Pharmacopoeia, 2015). Iridoid glycosides (mainly geniposide) and crocetin derivatives (crocins) are the two major active constituents in gardenia fruits (Fig. 1), and their components, characteristics, and activities have been investigated by many researchers (Han et al., 2015; Li et al., 2016; Li et al., 2017).

Numerous studies have demonstrated that geniposide and its aglycone genipin possess significant biological activities on inflammatory processes. A previous study suggested that genipin possessed strong anti-inflammatory activity, as demonstrated by the results of carrageenan-induced rat paw edema, carrageenan-induced air pouch formation, and measurement of NO content in the exudates (Koo et al., 2006). Another study indicated that geniposide exerts anti-inflammatory activity in lipopolysaccharide (LPS) stimulated primary mouse macrophages, and the same effect was observed in a mouse model (Fu et al., 2012). Moreover, anti-inflammatory and immunoregulatory effects of geniposide were observed, and these activities were exerted through inducing Th17 cell immune tolerance and enhancing Treg cell-mediated activities by down-regulating the expression of p-JNK (Dai et al., 2014). Furthermore, another investigation also revealed that geniposide exerted its anti-inflammatory effect by regulating TLR4 expression, which affected the downstream nuclear factor kappa B (NF- κ B) and mitogen-activated protein kinase (MAPK) signalling pathways (Song et al., 2014).

On the other hand, crocins, which are representative carotenoid compounds, have drawn considerable attention not only because they are a natural food colorant but also due to their great potential in medicine. A number of investigations have shown that crocetin derivatives (crocins) exert significant anti-inflammatory activity. A previous study indicated that crocin administration could reduce

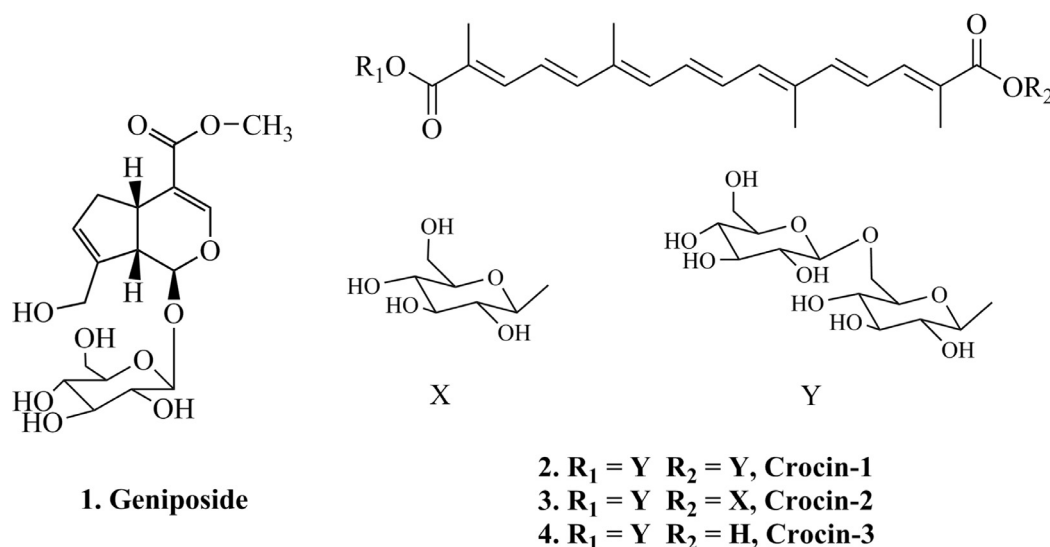


Fig. 1. Structures of geniposide and crocins.

inflammation-driven tissue impairment by decreasing the serum levels of pro-inflammatory cytokines TNF- α and IL-6, while increasing the level of anti-inflammatory cytokines IL-10 (Yang and Dong, 2017). Another investigation demonstrated that crocin possesses a significant protective effect against allergic asthma progression, which was associated with down-regulation of inflammatory cytokine expression (Yosri et al., 2017). In addition, another study implied that crocin inhibited IL-1 β -induced activation of matrix metalloproteinases, which likely play a pivotal role in the process of osteoarthritis (Ding et al., 2013). Furthermore, another paper indicated that crocin effectively suppressed degeneration-related inflammation in vitro and ex vivo (Kang et al., 2015). These data imply that crocin may possess obvious anti-inflammatory effects and could be one of the active ingredients in *G. jasminoides* Ellis or in the inflammation-modulating herbal remedy *Crocus sativus* L.

Gardenia has been used in many inflammation-related disease therapies. As mentioned above, its main components iridoid glycosides and crocetin derivatives possess significant biologic activities in inflammatory processes. Which components are responsible for the anti-inflammatory properties of gardenia, however, have remained to be investigated. To the best of our knowledge, the systematic anti-inflammatory effect of gardenia has not been well investigated. Therefore, the aims of the current study were designed for the first time to purify total iridoid glycosides (TIG) and total crocins (TC) from *G. jasminoides* Ellis, to determine its main chemical constituents, and to perform animal studies to comparatively evaluate its anti-inflammatory activities in Freund's complete adjuvant (FCA)-induced arthritis in rats. Adjuvant arthritis (AA) is a model of experimental arthritis induced by injection of FCA. Limb swelling, inflammatory cell infiltration, proliferative synovitis, and erosion of bone and cartilage structure are clinical findings common to human arthritis and adjuvant-induced arthritis in rats. In addition, AA in rats has similar histological and immunological characteristics to RA in humans and is therefore a useful assay model for assessing proliferation-inhibiting therapies for RA. Therefore, an AA rat model was applied to evaluate the biological activities of target components in the current investigation. Overall, the current investigation sheds light on reasonable explanations of the anti-inflammatory active components of this herbal remedy.

Material and methods

Reagent and materials

The dried *Gardenia jasminoides* Ellis fruits were purchased from Zhuhai, Guangdong Province, in November 2016, and identified by Yang Chen. The voucher specimens were deposited at the Department of Pharmaceutical Sciences, Zunyi Medical University, Zhuhai Campus, China. References geniposide, crocin-1, crocin-2 and crocin-3 (Fig. 1), were isolated by high-speed countercurrent chromatography (HSCCC) according to our previous study (Wang et al., 2015) and the resulting purities were 99.2%, 98.3%, 99.0% and 98.1%, respectively, according to the ultra-high-performance liquid chromatography (UHPLC) peak-area percentage. Diclofenac sodium (DS) was purchased from Renhe Pharmaceutical Co. (Beijing, China).

Extraction and separation

Separation and preparation of TIG and TC were conducted according to our previous study with minor modification (Chen et al., 2008). The dried ripe gardenia fruits (2.0 kg) were ground to a coarse powder and extracted with 20 l of 40% ethanol (v/v) by cold percolation. After concentration by a rotatory evaporator, the extract was dissolved in water and then subjected to column chromatography on HPD-100 macroporous resin and, eluted with water containing increasing amounts (0, 25, 40, 60%, v/v) of ethanol. The 25% alcohol-eluted fraction was evaporated to a water solution by a rotatory

evaporator, and then lyophilized to dryness to obtain TIG. In addition, TC was obtained from the 60% alcohol-eluted fraction by the same means.

UHPLC-DAD-MS and UV-Vis analysis

The UHPLC equipment was an Agilent 1290 Infinity System, equipped with a binary pump (G4220A) solvent manager, a DAD (G4212A) detector, a Hip sampler (G4226A), and a reversed phase SB C₁₈ column (2.1 mm $\times$ 50 mm, 1.8 μ m). The mobile phase consisted of A (acetonitrile) and B (0.5% formic acid-water). Gradient elution was programmed as follows: from 0 to 4 min, 5% to 10% A; 4–7.0 min, linear increase from 10% to 20% A; 7.0–9.0 min, linear increase from 20% to 33% A; 9.0–15.0 min, linear decrease from 30% to 40% A; 15.0–18.0 min, linear increase from 40% to 50% A; and 18.0–20.0 min, isocratic for 5% A. The flow rate was 0.3 ml/min while chromatograms were recorded at 440 nm for TC and 238 nm for TIG. The target peaks obtained by UHPLC were identified by their UV and HRESIMS spectra. MS analysis of the target peak was performed by using an ESI-Q-TOF mass spectrometer (Agilent 6530 Series, Agilent Technologies, Santa Clara, CA, USA) in positive ionization mode. The parameters for the Agilent Jet Stream Technology were a superheated nitrogen sheath gas temperature of 350 °C and a flow rate of 12 l/min. The electrospray conditions were as follows: capillary voltage: 4000 V; nebulizer: 20 psi; drying gas: 10 l/min; gas temperature: 350 °C; skimmer voltage: 60 V; octapole RFPeak: 750 V; and fragmentor voltage: 135 V. The data recorded were processed with a mass profiler (Agilent, Santa Clara, CA, USA) with accurate mass application specific additions based on Agilent MSD TOF software. A second orthogonal sprayer with a reference solution was used for continuous calibration with the following reference masses: 121.0509 and 922.0098 m/z. Spectra were acquired over the m/z 50–1000 range at a scan rate of 1 s per spectrum.

Furthermore, the UV-Visible spectra of TIG and TC were measured using a UV2600 UV/Visible spectrometer (Shimadzu, Japan) and compared with the spectra of geniposide and crocin-1.

Animals

Male specific pathogen-free (SPF) Sprague-Dawley rats (250–270 g) were obtained from the Experimental Animal Center of Guangdong Province (Guangzhou, China). The animals were housed at 25 $\pm$ 2 °C under a 12-h light/12-h dark cycle with ad libitum access to food and water. All animal experimental procedures were approved by the Animal Care and Use Committee of Zunyi Medical University (ethics approval (2014)2–043), and all efforts were made to minimize animal suffering and to reduce the number of animals used.

Arthritis induction and drug treatments

After acclimation for 1 week, the animals were randomly divided into four groups comprising six rats ($n = 6$) in each group as follows:

- Group I: Healthy control.
- Group II: Adjuvant arthritis control.
- Group III: Adjuvant arthritis rats with 160 mg/kg TIG treatment.
- Group IV: Adjuvant arthritis rats with 160 mg/kg TC treatment.
- Group V: Adjuvant arthritis rats with 5 mg/kg DS treatment.

Arthritis was induced by a single intra-dermal injection of 0.1 ml FCA into the foot-pad of the right hind paw. The adjuvant contained heat-killed mycobacterium tuberculosis (10 mg) in paraffin oil (1 ml). TIG and TC were prepared as aqueous suspensions in water. All rats except controls underwent drugs treatment by gavage for 14 consecutive days from days 15–28 post-arthritis induction. Healthy control and arthritic control rats received vehicle (water) by gavage during the 14 days of treatment.

Radiological analysis of ankle joints

On day 28, rats were anaesthetized, and X-ray radiographs of the adjuvant injected hind paws were taken (X-ray Diagnostic System Ysio, Siemens AD, Germany) to determine any bone changes. Radiographic measurement of the hind paws was performed at 52 kV peak, 5.0 mAs, with an exposure time of 1 s. To evaluate the degree of soft tissue swelling/joint destruction and periarticular erosive changes, radiographs were also scored according to previous studies (Gowayed et al., 2015). For soft tissue swelling, radiographs were scored on a scale of 0–4 (0 = normal with no soft tissue swelling, 1 = minimal soft tissue swelling, 2 = mild to moderate soft tissue swelling, 3 = moderate soft tissue swelling, 4 = marked to severe soft tissue swelling). For the degree of joint destruction and periarticular erosive changes, radiographs were also scored from 0–4 (0 = no joint destruction, 1 = minimal joint destruction, 2 = mild to moderate joint destruction, 3 = moderate joint destruction, 4 = marked to severe joint destruction with evident periarticular erosive changes). All scores were assessed by an observer blinded to the experimental groups.

Measurement of joint diameter

Joint diameters were measured using a digital Vernier caliper (SMCT, China) on day 0 before FCA injections and thereafter on days 1, 4, 8, 12, 14, 16, 18, 20, 22, 24, 26 and 28.

Histopathological and immunohistochemical examination of ankle joints

On day 28, the right hind ankle joints of the rats were removed and immediately fixed in 4% paraformaldehyde for histopathological and immunohistochemical analysis. The ankle joints were then decalcified in 10% ethylenediaminetetraacetic acid (EDTA) for 30 days at 4 °C, dehydrated in a gradient ethanol series, washed twice with xylene, and embedded in paraffin. Finally, they were cut into sections of approximately 5 µm in thickness, stained with hematoxylin-eosin (H&E) and observed by light microscopy. According to a previous report with modification (Goff et al., 2009), briefly, the ankle joints were histopathologically analysed for synovial proliferation, inflammation, articular cartilage destruction and erosion, synovial lining cell

enlargement, synovial vessel thrombus, focal cellulose deposition and papillary tissue (scales 0–3: 0, no detectable changes; 1, mild; 2, moderate; 3, severe). The pathological evaluations were performed randomly by a pathologist who was blinded to the groups of the specimens.

Furthermore, we applied antibodies against rat TNF-α and TGF-β1 and analysed the amounts of antibodies in ankle joint tissues. Positive expression of antibodies was indicated by brown or yellow colour, and the colour intensity was correlated with the amount of antibodies.

Statistical analysis

The data were analysed using SPSS.19 (SPSS Inc., Chicago, USA), expressed as the means ± standard deviation (SD) and statistically analysed by one-way analysis of variance (ANOVA) test. The Tukey multiple comparison test was then performed to determine differences between groups. In addition, the expression levels of target proteins were analysed by the average integrated optical density using IPWIN60 software (Media Cybernetics, Inc., Rockville, MD, USA). A *P*-value < 0.05 was considered statistically significant.

Results

UHPLC-DAD-MS and UV–Vis determination of two fractions from *Gardenia jasminoides* Ellis

In the current investigation, separation and purification of 2.0 kg gardenia fruits yielded two main fractions, one was a brown amorphous powder and the other was a deep red powder. Moreover, UHPLC-DAD-MS was further conducted to clarify the compositions of these two fractions. Under the LC-MS conditions used in the current study, the data showed that the predominant component in the brown amorphous powder was geniposide, whose major molecular ion was confirmed by $[M + Na]^+$ at 411.1262 (Calcd for $C_{17}H_{24}O_{10}Na$ $[M + Na]^+$, 411.1286) (Fig. 2). In addition, determination of the deep red powder revealed three major peaks which were subsequently characterized by MS spectra to be crocin-1 (999.3680, Calcd for $C_{44}H_{64}O_{24}Na$ $[M + Na]^+$, 999.3734), crocin-2 (837.3152, Calcd for $C_{38}H_{54}O_{19}Na$ $[M + Na]^+$, 837.3200) and crocin-3 (675.2623, Calcd for $C_{32}H_{44}O_{14}Na$

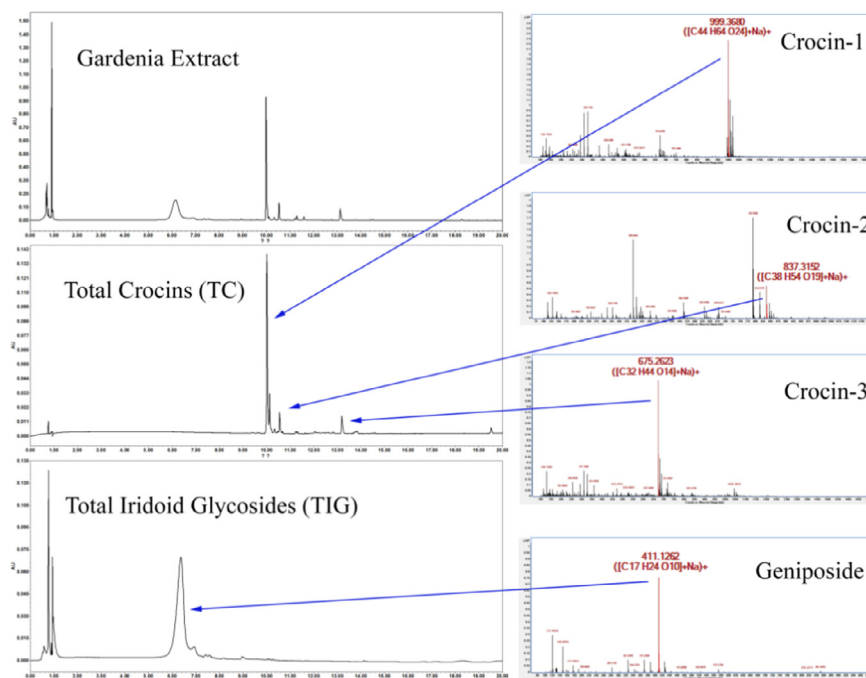


Fig. 2. LC-MS chromatograms of TIG and TC.

[M + Na]⁺, 675.2671) (Fig. 2). Furthermore, these four components were further identified by comparing their retention times with those of the reference geniposide, crocin-1, crocin-2 and crocin-3 samples, respectively. On the other hand, UV–Vis determination of the two fractions, in comparison with the reference geniposide and crocin-1, further confirmed that these two amorphous powders were TIG and TC (Fig. 3), respectively. By comparing the areas of analytes in chromatograms with those of the four references, geniposide, crocin-1, crocin-2 and crocin-3 were determined to be 53.6%, 37.2%, 3.3% and 4.6% in corresponding fractions, respectively.

Apart from these four components, as shown in Fig. 2, similar UV spectra of some small peaks with geniposide and crocins indicated that these two fractions also contained other similar low-content derivatives. Thus, these other derivatives should be determined together with the four main components. Therefore, the concentrations of TIG and TC were determined with a UV spectrophotometer, in comparison with references geniposide and crocin-1, respectively. Comparisons of absorptions of TIG and TC at 238 nm and 440 nm, with those of geniposide and crocin-1 indicated that TIG and TC contained 58.2% of total iridoid glycosides and 54.7% of total crocins, respectively (Table 1). In general, the UHPLC chromatograms of the crude extract and two fractions implied that main iridoid glycosides and crocins were effectively enriched in the two fractions (Fig. 2), which were subsequently subjected to *in vivo* assays.

Effects of TIG and TC on paw swelling and diameters

Hind paw swelling manifests both inflammatory and arthritic alterations in AA rats. The arthritic rats showed significant macroscopic signs of severe arthritis, such as swelling, redness and deformity, in the right hind paw and ankle joint (Fig. 4) if compared with the healthy control group. The right hind paw diameters were determined to assess the extent of AA induced by Freund's complete adjuvant. As shown in Fig. 4, a significant increase in paw diameters was observed in the adjuvant-treated group on days 16, 18, 20, 22, 24, 26 and 28 compared with those in the control group. However, paw diameters strikingly decreased after TIG (160 mg/kg), TC (160 mg/kg) and DS (5 mg/kg) treatments in comparison with those of the model group (all $p < 0.05$) (Fig. 4).

Radiological examination

Fig. 5 shows normal bone matrix in healthy control rats, whereas adjuvant-induced arthritis resulted in bone deformation in AA rats indicated by severe joint erosion and cystic enlargement of bone, with associated soft tissue swelling. Treatment with TIG (160 mg/kg) significantly ameliorated both paw swelling and bone destruction, and reduced radiological bone destruction score (Fig. 5). DS treatment also markedly decreased radiological bone destruction score ($p < 0.05$). TC (160 mg/kg), similar to TIG (160 mg/kg), mitigated the changes in bone structure manifested as an absence of erosive bone changes and soft tissue swelling. However, significantly stronger anti-arthritis activity was revealed for TIG ($p < 0.05$) based on radiological bone destruction scores, while TC induced no significance difference ($p > 0.05$) (Fig. 5).

Histological assessment

Histological assessments of the joint provided direct evidence of protective effects of TIG and TC on adjuvant-induced arthritis. Histopathological evaluation of healthy control joints showed an intact joint structure with no cartilage or bone destruction (Fig. 6). After immunization with FCA, there was a significant increase in joint inflammation in all experimental groups compared with the healthy control group. The joints of AA rats presented prominent synovial hyperplasia, infiltration of inflammatory cells and marked articular

cartilage destruction and erosion in comparison with the healthy control group (Fig. 6). Moreover, the synovial lining cell layer was substantially enlarged to 4–5 layers, synovial vessels were dilated, which led to thrombus formation, and focal cellulose deposition was observed in AA rats. However, the animals that received TIG (160 mg/kg) and DS (5 mg/kg) demonstrated minimal synovial hyperplasia (Fig. 6) and mitigation of joint damaged when compared with the arthritic control rats (both $p < 0.01$). Although the synovial liner extending into the articular cavity and papillary tissue were occasionally observed after administration of TC (160 mg/kg), this group showed significant recovery and reduced joint damage. Like the TIG-treated group, the histological damage score of the TC-treated group corresponded to $p < 0.01$. These results suggested that the main components of gardenia had therapeutic effects against arthritis in this assessment, and these two components may possess comparable anti-inflammatory properties.

Moreover, the spleen is the largest secondary lymphoid organ in rats and houses peripheral B and T cell compartments. Therefore, histopathological measurements of the spleens were carried out to evaluate the inflammatory reactions of the animals. In AA rats, significant red pulp hyperplasia and congestion were observed, and the boundaries between white pulp and red pulp were unclear (Fig. 7). In contrast, TIG-, TC- and DS-treated rats exhibited only minimal hyperplasia of the red pulp and minimal pathological changes (Fig. 7). In addition, the thymus can generate mature T cells, and is a primary lymphoid organ. Therefore, histopathological measurements of the thymus were also conducted to evaluate inflammatory reactions. As shown in Fig. 7, the peripheral cortex of the thymus was also congestive, while mild oedema and many macrophages were observed in the model group. However, the animals that received TIG, TC and DS demonstrated significant improvement compared with the arthritic control rats, and TC seems to have comparable potential to TIG in both organs.

Immunohistochemistry analysis

To examine if the observed anti-arthritis properties of TIG and TC were due to the deactivation of TNF- α and TGF- β 1 protein expression levels, we further examined the expression levels of these two proteins by immunohistochemistry staining of synovial tissues of healthy control and experimental rats. As shown in Fig. 8, the brown areas in the photographs show the homologous expression of TNF- α and TGF- β 1 antibodies. Moreover, the expression levels of TNF- α and TGF- β 1 were analysed by the average integrated optical density using IPWIN60 software. As shown in Fig. 8, there was little TNF- α and TGF- β 1 in the healthy control group, while these two proteins were strongly expressed in the synovial tissue of arthritic rats if compared with healthy control

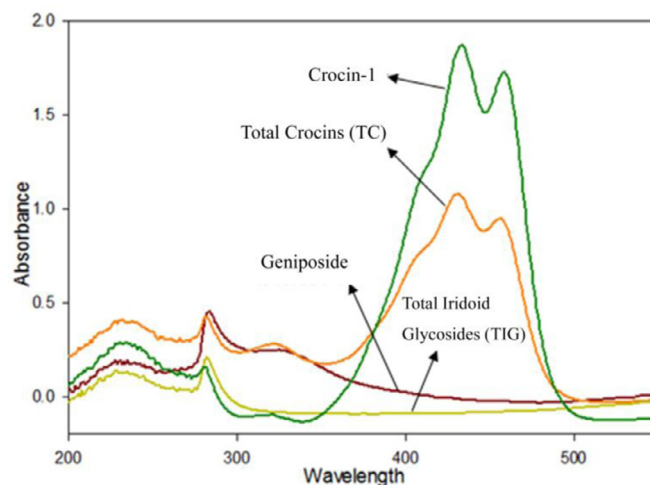


Fig. 3. UV–Vis spectra of TIG and TC.

Table 1
Percentages of main components in crude extract and two fractions from gardenia.

Fractions	geniposide	crocin-1	crocin-2	crocin-3	total amounts
Crude extract	19.4%	7.1%	0.93%	1.08%	–
TIG	53.6%	–	–	–	58.2%
TC	–	37.2%	3.3%	4.6%	54.7%

rats. However, TIG treatment (160 mg/kg) significantly suppressed expression of TNF- α ($p < 0.01$) and TGF- $\beta 1$ ($p < 0.01$) proteins in the synovial tissue of arthritic rats. In addition, DS treatment markedly decreased expression levels of TNF- α ($p < 0.01$) and TGF- $\beta 1$ ($p < 0.01$) proteins. In contrast, TC also decreased the amount of TNF- α ($p < 0.05$) and TGF- β ($p > 0.05$) (Fig. 8). The current work indicated that both TIG and TC from gardenia exert anti-inflammatory properties by modulating these protein expression levels. However, comparison of data revealed that iridoid glycosides seemed to have stronger activity and

likely play a more important role in the anti-arthritis properties of gardenia.

Discussion

Geniposide (and other iridoid glycosides) has been widely considered to be the major component of *G. jasminoides* Ellis, and is a standard chemical for quality evaluation of gardenia in current China Pharmacopoeia (National Commission of Chinese Pharmacopoeia, 2015). Indeed, numerous reports have demonstrated that this component is responsible for the anti-inflammatory properties of this herbal remedy (Koo et al., 2006). On the other hand, another investigation indicated that crocetin, the aglycone of crocins, is probably the main anti-inflammatory component of gardenia (Hong and Yang, 2013). According to previous reports (Chen et al., 2010) and the current study (Table 1), lower yet significant amounts of crocetin derivatives (crocins) are contained in gardenia. Additionally, crocins have been shown to possess significant anti-inflammatory effects in different

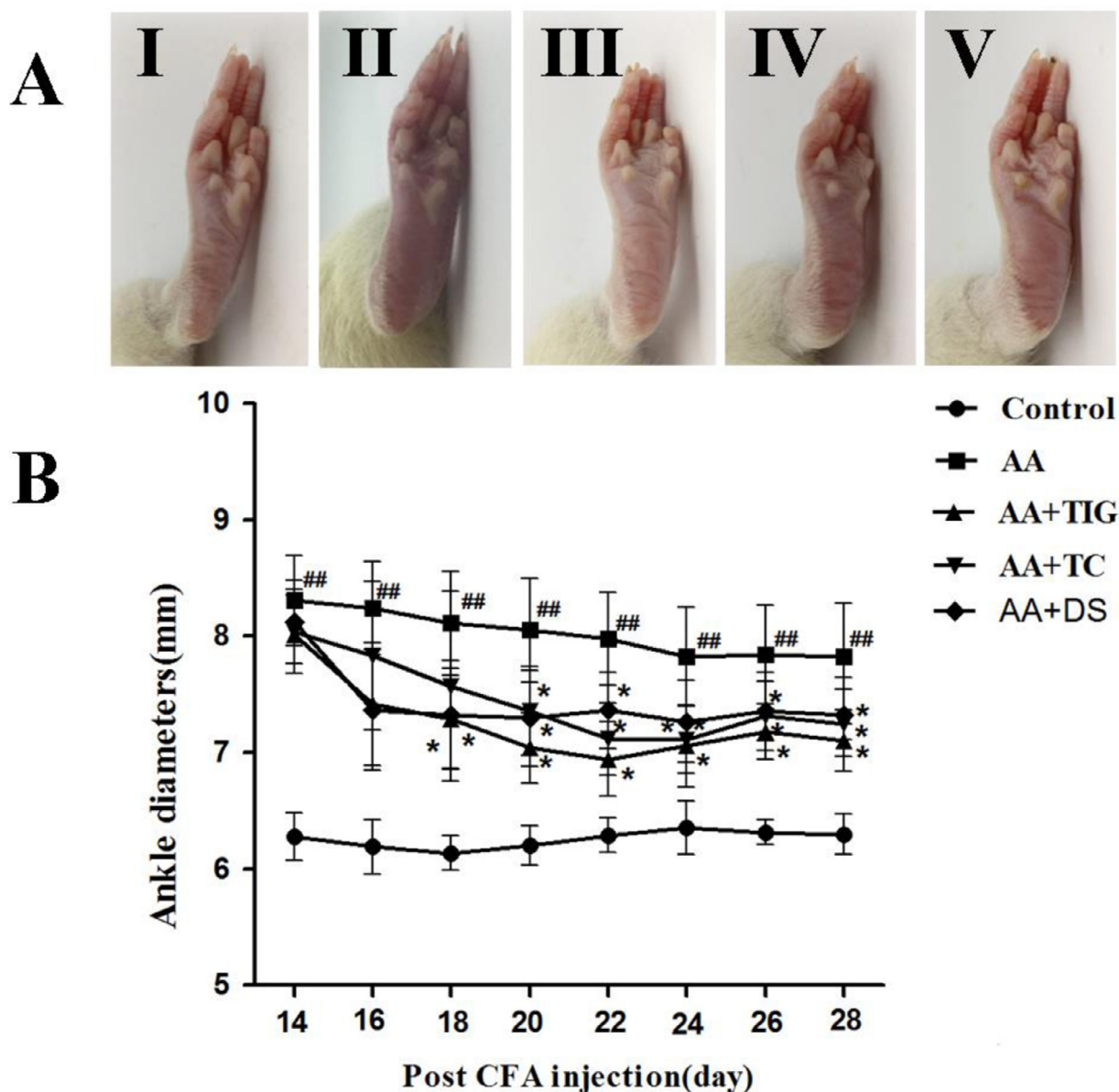


Fig. 4. (A) Representative photographs of the right hind paw of healthy control and arthritic rats on day 28 after adjuvant injection. (I) Healthy control. (II) Arthritic control. (III) TIG 160 mg/kg treated. (IV) TC 160 mg/kg treated. (V) DS 5 mg/kg treated. (B) Effect of TIG and TC (160 mg/kg) on hind paw diameters in adjuvant arthritic rats. Data presented as mean $\pm$ S.D., and are compared with healthy control, ## at $P < 0.01$, and are compared with AA groups, * at $P < 0.05$.

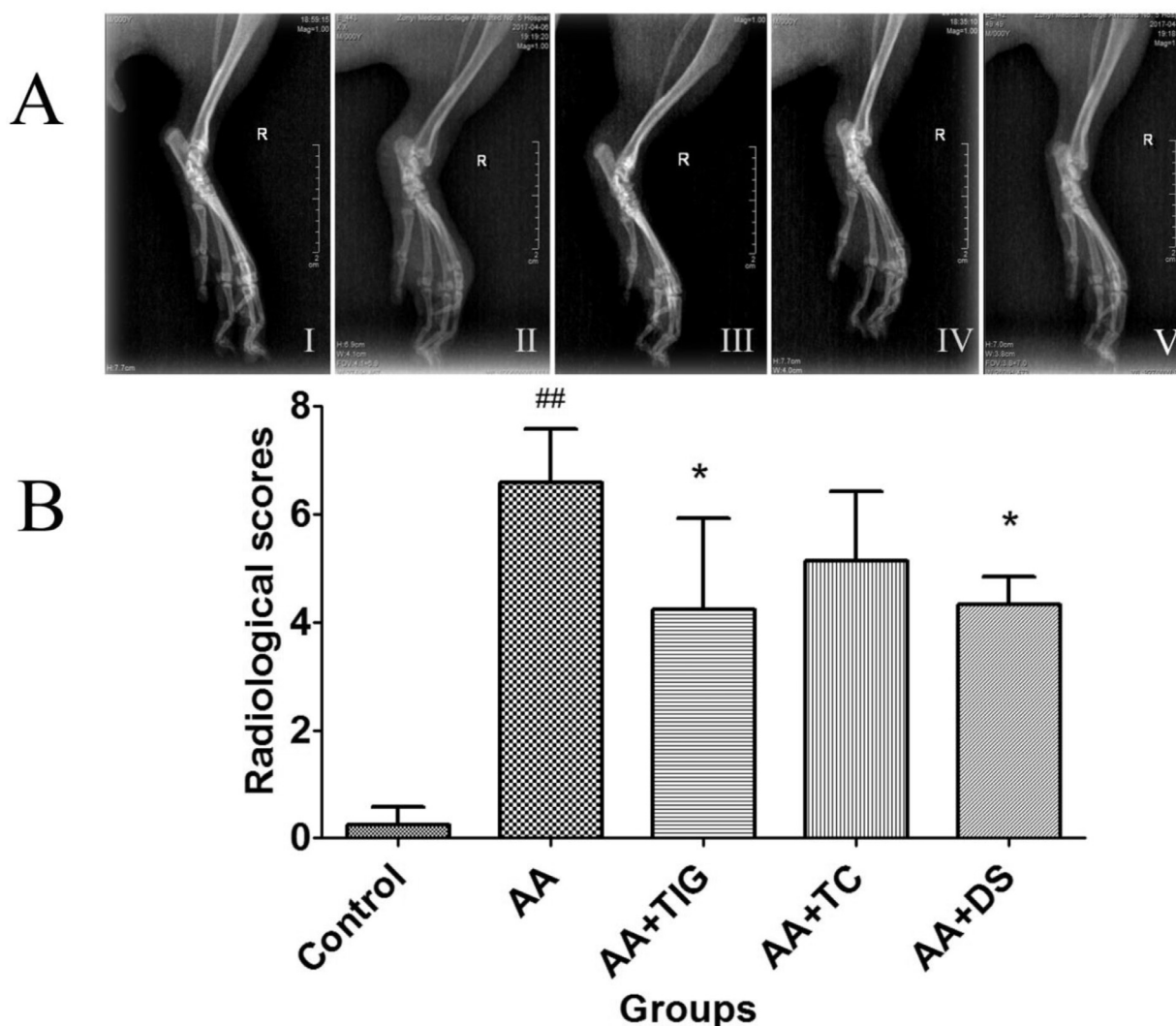


Fig. 5. (A) Radiological images of ankle joints. (I) Healthy control. (II) Arthritic control. (III) TIG 160 mg/kg treated. (IV) TC 160 mg/kg treated. (V) DS 5 mg/kg treated. (B) Radiological scores of ankle joints. Data presented as mean $\pm$ S.D., and are compared with healthy control, ## at $P < 0.01$, and are compared with AA groups, * at $P < 0.05$.

models. Therefore, the current investigation aimed to provide a reasonable explanation for the anti-inflammatory active components of gardenia using phytochemical separation techniques.

Separating components from herbal medicines is a challenging task. The whole extracts of crude drug are routinely subjected to *in vivo* assays without further purification. However, studies relying on this type of protocol cannot reveal which components are responsible for biological activity of whole herb. To uncover the main role of each herbal medicine, separation of main components of herbs could be carried out prior to further *in vivo* investigation. However, processes of separation of compounds with high purity are tedious and time-consuming and require multiple chromatographic steps, which results in lower recovery and higher costs. Therefore, separation of components from herbs could be conducted by fractionation, during which the same derivatives could be enriched in one fraction with a high yield. In the present work, TIG and TC were prepared by an HPD-100 macroporous resin, and UHPLC-DAD-MS measurements confirmed that the main chemicals in gardenia were effectively enriched into two fractions, namely, TIG and TC (Fig. 2). In addition, the contents of these main components TIG and TC were determined by UV-Vis determination, and the data showed that iridoid glycosides and crocins are main components of these two fractions (Table 1).

RA is a chronic systemic disease with clinical manifestations of multi-joint damage, leading to chronic pain, joint deformity and functional disability (Soeken et al., 2003). The typical preclinical experimental models of RA include the adjuvant-induced arthritis model and the collagen-induced arthritis model (Tong et al., 2015; Wu et al., 2016). The arthritic etiology of AA and RA exhibits common immunological and pathological features, including the involvement of inflammatory mediators, immune dysfunction and bone erosion (Yu et al., 2011). Additionally, the rat AA model is an easy, reliable and reproducible experimental model of polyarthritis that can be rapidly developed. It has been commonly used for preclinical evaluations of anti-arthritic drugs due to its similar pathology to human RA (Bendele, 2001; Wang et al., 2016). Therefore, the AA rat model induced by FCA injection has been broadly used as an experimental animal model for investigating pathogenesis of RA and for screening new therapeutic methods to cure rheumatoid diseases (Bhalekar et al., 2016). Data from previous investigation indicated that geniposide promoted recovery of arthritis and inhibited colonic inflammatory damage in AA rats by decreasing the expression levels of TNF- α , IL-1 and IL-6, increasing the production of IL-10, and inhibiting the expression of phospho-p38 related proteins in fibroblast-like synoviocytes (Chen et al., 2015). Moreover, data have shown that this component might exert its anti-

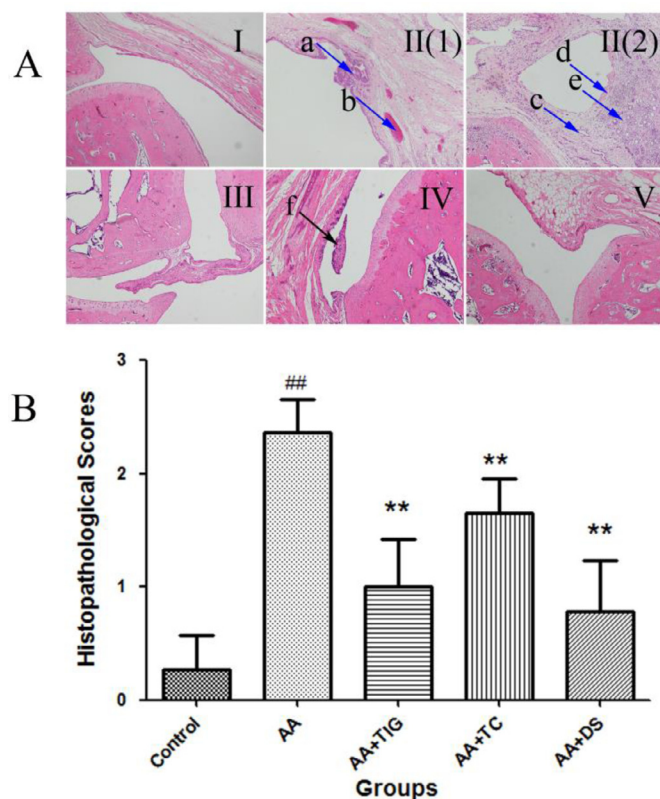


Fig. 6. (A) Histopathological photos and scores of ankle joints stained with H&E. ($\times 100$) (I) Healthy control. (II) Arthritic control. (III) TIG 160 mg/kg treated. (IV) TC 160 mg/kg treated. (V) DS 5 mg/kg treated. The blue arrow (a) represents synovial lining cells enlarged to 4–5 layers; the blue arrow (b) represents synovial vessels dilated with thrombus; the blue arrow (c) represents synovial vessel dilation; the blue arrow (d) represents focal cellulose deposition; the blue arrow (e) represents infiltration of inflammatory cells; the black arrow (f) represents papillary tissue. (B) Histopathological scores of ankle joints. Data presented as mean $\pm$ S.D., and are compared with healthy control, ^{##} at $P < 0.01$, and are compared with AA groups, ^{**} at $P < 0.01$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

inflammatory and immunoregulatory effects through inducing Th17 cell immune tolerance and enhancing Treg cell-mediated activities by down-regulating the expression of p-JNK (Dai et al., 2014). Further *in*

vitro study indicated that geniposide induces AA fibroblast-like synovocyte (FLS) apoptosis through regulating apoptosis-related gene expression levels and inhibiting ERK signal pathway (Li et al., 2016). On the other hand, PK data showing the distribution of geniposide in synovium tissue supported possible anti-inflammatory activity of geniposide in a rat AA model (Chen et al., 2014; Chen et al., 2015). Furthermore, another investigation indicated that crocin exerted therapeutic anti-inflammatory effects in an RA rat model, probably by reducing iNOS production and TNF- α , IL-1 β , and IL-6 serum content (Xiang et al., 2017).

To investigate the effects of the main components of gardenia in an AA rat model, morphological observation, radiological analysis, histopathological study and immunohistochemistry were performed in the current investigation. Consistent with the abovementioned findings, the results of the current study generally demonstrated that TIG and TC from gardenia alleviated the severity of AA. Paw swelling is an index of arthritic severity, and reduced ankle diameters have been used as an index for measuring the anti-arthritic activity of various drugs (Gowayed et al., 2015; Kumar et al., 2016; Lad and Bhatnagar, 2016). According to morphological observation, the AA model rats showed secondary inflammation and increased ankle diameters. However, TIG and TC at a dosage of 160 mg/kg significantly decreased the FCA-induced ankle diameters when compared with the arthritis group (both $p < 0.05$) (Fig. 4). In addition, bone destruction, a common feature of AA, has been conventionally evaluated by radiological analysis in anti-arthritic drugs investigation (Cunha et al., 2016; Gowayed et al., 2015; Ito et al., 2016). The results of our study showed that TIG treatment (160 mg/kg) significantly ameliorated both paw swelling and bone destruction, and reduced the radiological bone destruction score ($p < 0.05$) while TC treatment had no such significant effects ($p > 0.05$).

As a systemic inflammatory disorder, RA is characterized by synovial hyperplasia, chronic inflammation, and progressive articular destruction of cartilage and bone. RA mainly targets the synovium and joints as the primary sites of the inflammatory process, which is reflected by histopathological assay. Therefore, histopathology is a prominent method used to evaluate the morphologically pathological characterizations of RA (Obiri et al., 2014). In the current investigation, histopathological analyses of joints further demonstrated the anti-inflammatory and protective effects of TIG and TC against joint impairment, including synovial hyperplasia, articular cartilage destruction and erosion, infiltration of inflammatory cells, synovial lining cells enlargement, synovial vessels dilation with thrombus formation and focal cellulose deposition (both $p < 0.01$) (Fig. 6). In addition, spleen normally filters out and destroys old and damaged blood cells and removes antigens by phagocytosis, thus playing a key role in preventing

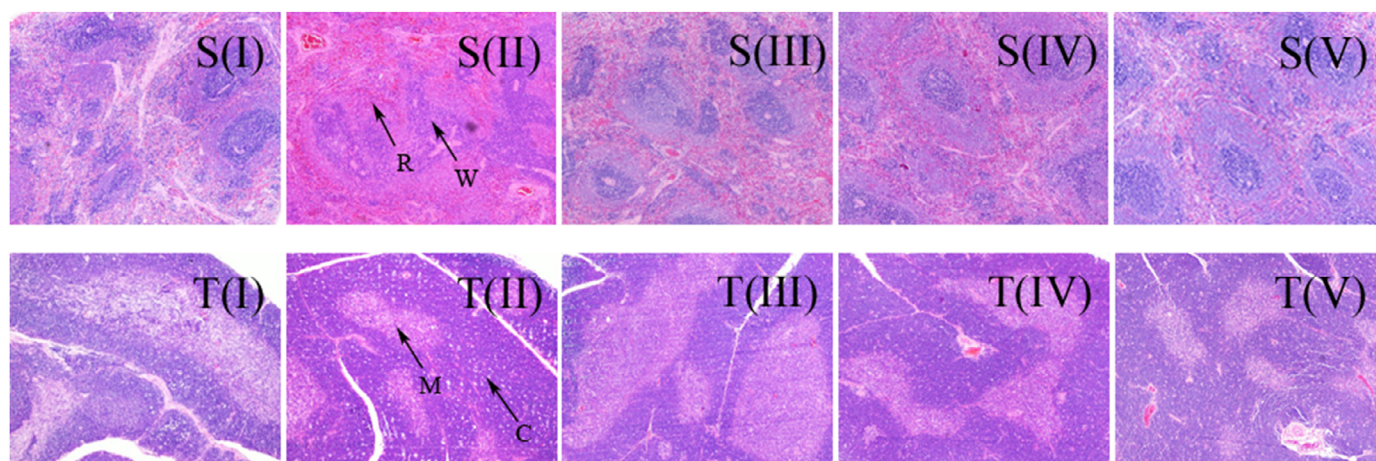


Fig. 7. Histopathological photos of spleens (S) and thymuses (T) stained with H&E. ($\times 100$) (I) Healthy control. (II) Arthritic control. (III) TIG 160 mg/kg treated. (IV) TC 160 mg/kg treated. (V) DS 5 mg/kg treated. The histology section shows the white pulp (W), the red pulp (R), central medulla (M) and peripheral cortex (C). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

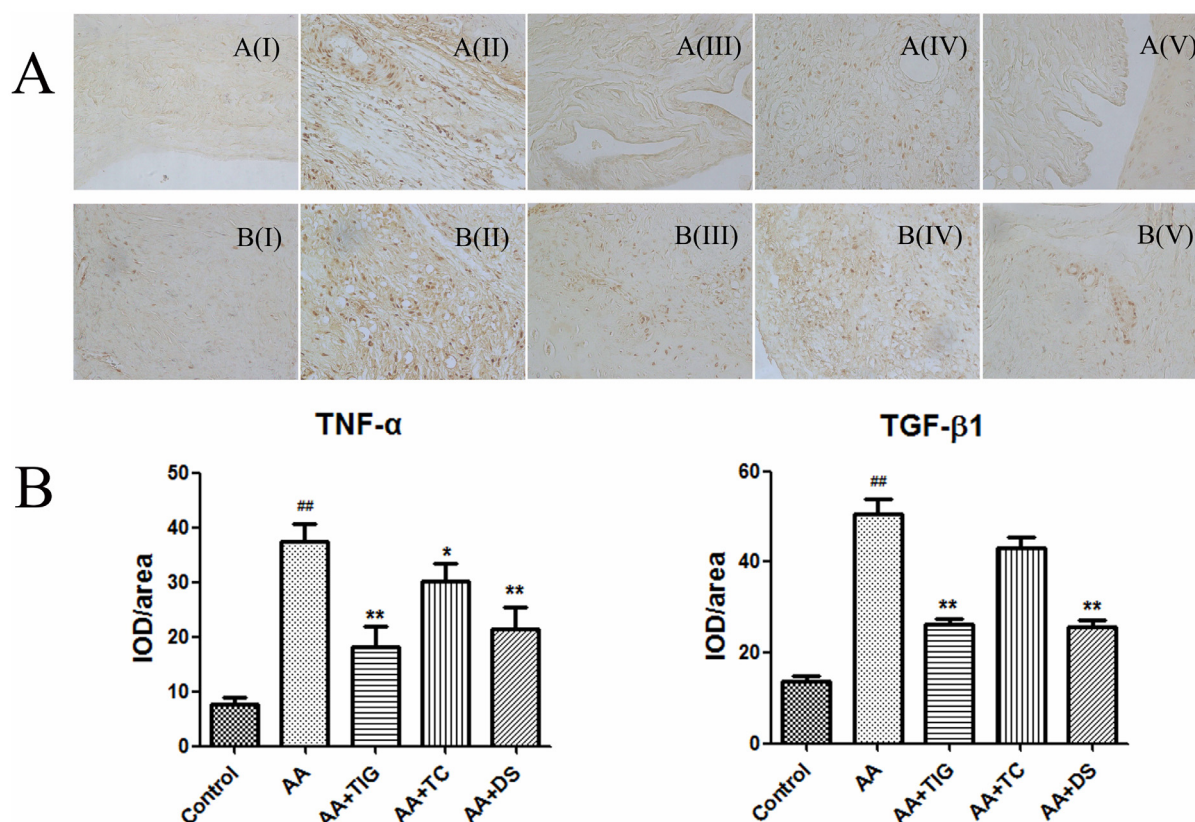


Fig. 8. Effects of TIG and TC on the expression of TNF- α and TGF- β 1 in joint synovium. (A) AA rats' synovial tissues and healthy control synovial tissues were tested by immunohistochemistry ($\times 400$). (I) Healthy control. (II) Arthritic control. (III) TIG 160 mg/kg treated. (IV) TC 160 mg/kg treated. (V) DS 5 mg/kg treated. (B) IOD values of ankle joints. Data presented as mean $\pm$ S.D., and are compared with healthy control, ^{##} at $P < 0.01$, and are compared with AA groups, ^{**} at $P < 0.01$, and are compared with AA groups, ^{*} at $P < 0.05$.

infection by producing white blood cells called lymphocytes and acting as a first line of defence against invading pathogens (Handy et al., 2002). Similar arguments apply to the thymus, which normally processes immature precursor T-lymphocytes (derived from bone marrow and initially located in the outer cortex of thymus) into the mature immunocompetent T cells of the medulla (Vos and Van, 1998). Our results clearly revealed that TIG and TC suppress inflammatory reactions in animals, as indicated by the mitigation of red pulp hyperplasia and congestion in the spleen, alleviation of white pulp congestion, and decreased numbers of macrophages in the thymus (Fig. 7).

Cytokines and mediators regulate a broad range of inflammatory processes that are implicated in the pathogenesis of RA. In response to infectious agents in RA, numerous pro-inflammatory cytokines and mediators are released and involved in the inflammatory response, which are responsible for the pain and damage to bone and cartilage that can lead to severe disability (McInnes and Schett, 2007; McKenna and Hunder, 1961). In addition, a complex regulatory network of cells and cytokines is directly implicated in many immunological processes of RA (McInnes and Schett, 2011; Siebert et al., 2015). It is well known that an imbalance between pro- and anti-inflammatory cytokines promotes initiation and progression of RA (McInnes and Schett, 2007). Particularly, TNF- α plays a significant role in the pathogenesis of RA by inducing the infiltration of immune cells and stimulating the release of matrix metalloproteinases (Brennan and McInnes, 2008). Furthermore, TNF- α is considered the most important pro-inflammatory cytokine and the dominant mediator of immune-mediated joint disease in RA patients and experimental models (Stolina et al., 2009). Therefore, a recent study suggested that pro-inflammatory cytokines such as TNF- α could be reliable biomarkers in RA (Staff, 2016). On the other hand, TGF- β 1 is one of the most powerful and widely distributed

profibrogenic mediators in the body. A previous research has indicated enhanced expression of TGF- β 1 protein in RA synovial, and it may be related with active pathological changes in RA synovia including synovial hyperplasia, inflammatory cell infiltration and granuloma formation (Dai et al., 2000; Szekanecz et al., 1995). In addition, data from other reports have shown that systemic inhibition of TGF- β 1 signalling can significantly reduce the severity of synovitis in RA animal models, likely by suppressing TGF- β 1-mediated synovial proliferation and endothelial angiogenesis (Sakuma et al., 2007; Wahl et al., 1993). Therefore, inhibiting these pro-inflammatory cytokines and mediators might be an effective strategy for the treatment of RA. To further investigate the mechanism of TIG and TC responsible for alleviating AA activities, we focused on TNF- α and TGF- β 1 in the current investigation. As shown in Fig. 8, immunization with adjuvant resulted in significant increases in TNF- α and TGF- β 1 proteins in the synovial tissue of arthritic rats compared with healthy control rats. However, administration of TIG (160 mg/kg) could suppress the expression of TNF- α ($p < 0.01$) and TGF- β 1 ($p < 0.01$) proteins in the synovial tissue of arthritic animals, while treatment with TC could only significantly decrease the amount of TNF- α ($p < 0.05$). These results suggest that the protective effects of TIG and TC on adjuvant-induced arthritis in rats may be partly attributed to regulation of TNF- α and TGF- β 1. However, data implied that iridoid glycosides may have stronger activity and probably play a more important role in suppressing TGF- β 1 protein.

Searching for novel, high-activity compounds is critical in natural products research (Savić et al., 2015). However, identifying the main components of the whole plant that are responsible for the efficacy of herbal medicines is a challenging task. Clues regarding main biological contributors of the corresponding herbal remedies could be revealed in the abovementioned studies. However, the contributing percentages of

natural products in a drug cannot be estimated mainly due to a lack of data pertaining to the contents of the main components in the herbal remedy. On the other hand, the relationship between the pharmacokinetic profile of each principle and its efficacy remains to be investigated. In the current investigation, the anti-inflammatory properties of the main chemicals from gardenia were compared. Although anti-inflammatory activity was observed in the TC-treated group, the results implied that the main anti-inflammatory contributor in gardenia is probably geniposide, considering the dramatically higher content of geniposide compared with the content of crocins (Chen et al., 2010) (Table 1).

As mentioned in the “Introduction”, numerous investigations have implied that gardenia, as well as the two main chemicals geniposide and crocins, have significant biologic activities in inflammatory processes. However, which components is the main contributor in this herbal remedy remain unknown. The current study showed that both TIG and TC from gardenia have anti-inflammatory properties, but the former components iridoid glycosides seem to possess more remarkable anti-inflammatory ability in the majority of parameters. With regard to the higher content of geniposide compared with that of crocins, iridoid glycosides (particularly geniposide) are therefore probably more important in evaluating the anti-inflammatory activity of *G. jasminoides* Ellis. In our study, TIG and TC from gardenia mitigated disease progression and RA symptoms, generally supported by the results of the chemical determination and the assessment of paw swelling, ankle diameter, radiology analysis, histology and immunohistochemistry of ankle joint tissues in arthritis rats.

In conclusion, fractionation of gardenia prior to further in vivo investigation could provide a reasonable explanation for the anti-inflammatory activities of the active components in this herbal remedy. Our series of experiments suggested that the main components of gardenia prominently relieved systemic AA severity. Our study shows for the first time that both TIG and TC from gardenia have anti-inflammatory properties. Overall, these experimental findings suggest that gardenia could be regarded as a potential therapeutic target for arthritis. However, with regard to the higher content of geniposide compared with the content of crocins in this herbal drug, TIG (mainly geniposide) seems to be primarily responsible for the anti-inflammatory property of gardenia. Taken together, this initial attempt demonstrated that TIG (mainly geniposide) is more important in evaluating the anti-inflammatory activity of *G. jasminoides* Ellis.

Conflict of interest

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

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